

Module 3: Applications of First-Order ODEs

Welcome to **Module 3**. Now that you have mastered the techniques for solving first-order differential equations, we can apply these skills to solve geometric and physical problems.

Differential equations are not just abstract mathematical constructs; they are the language used to describe the physical universe, from the motion of planets to the flow of electricity. In this module, we focus on a specific geometric application found in your course notes: **Orthogonal Trajectories**.

Module Learning Objectives

By the end of this module, you will be able to: 1. Understand the geometric relationship between families of curves. 2. Calculate the **Orthogonal Trajectories** of a given family of curves (curves that intersect at right angles).

Topic Index

Navigate through the applications covered in this module using the link below:

1. Orthogonal Trajectories

- **The Concept:** What does it mean for two families of curves to be orthogonal?
 - **The Method:** Using the negative reciprocal of the slope ($-\frac{dx}{dy}$) to find the perpendicular family.
 - *Includes step-by-step solved questions for algebraic and trigonometric curves.*
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Next Steps: Proceed to the topic [Orthogonal Trajectories](#) to see how differential equations define the shape of perpendicular curves.